

TITLE PAGE

- **Problem Statement ID – 25004**
- **Problem Statement Title- Image-based Breed Recognition for Cattle and Buffaloes of India**
- **Theme- Agriculture, FoodTech & Rural Development**
- **PS Category- Software**
- **Team ID- SIH**
- **Team Name- (InVisionX)**



IMAGE BASED BREED RECOGNITION SYSTEM FOR CATTLE AND BUFFALOES IN INDIA

❖ Proposed Solution (Idea/ Solution/ Prototype) :-

Present **Label Blend**, a **synthetic data generation tool** that **automatically creates large volumes of labeled images** (Bounding Box / Segmentation-based datasets) of **cattle and buffalo** under varying **poses, environments, and lighting conditions**. This **augments and enriches datasets** for **robust breed classification**.

How It Addresses the Problem:

- ✓ **Overcomes limited and inconsistent training data**
- ✓ **Ensures breed recognition accuracy under real-world field conditions**
- ✓ **Integration with Bharat Pashudhan App (BPA): feeding high-quality synthetic datasets into model training pipelines**
- ✓ **Improves AI model generalization across diverse cattle and buffalo breeds**

Innovation & Uniqueness:

- **Customizable environment and pose generation** → enhances dataset diversity
- **Auto-labeled synthetic data** → accelerates **AI/ML development lifecycle**
- **Scalable solution** → adaptable across **indigenous and crossbred variations**
- Strengthens **Computer Vision, Deep Learning, and Image Recognition** performance in livestock management



TECHNICAL APPROACH

Technologies we Use in this Project:

- **Python** for (Backend & AI/ML, Data Processing, Scripting),

Frameworks:

Django, REST-API

JavaScript (Frontend & Web Development),

Spring Framework, SQL

PySide (Open Source GUI Development)

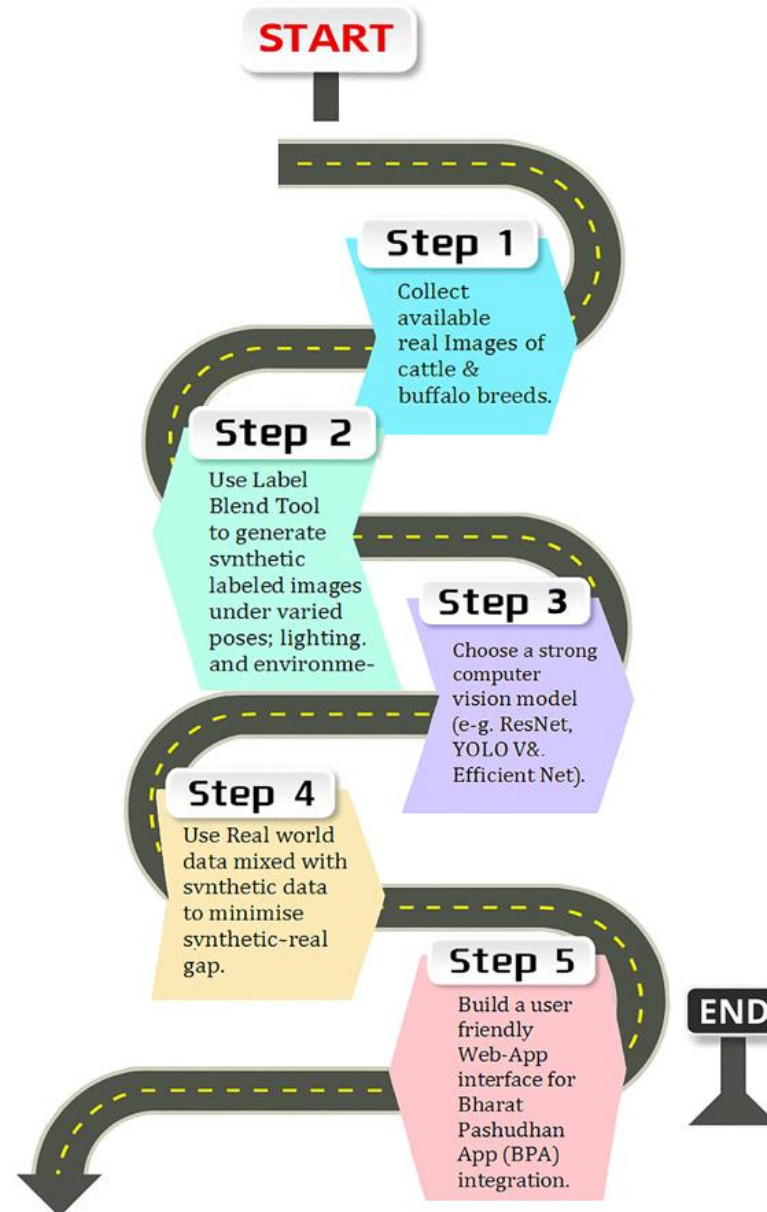
Qt Framework (Cross-platform UI/UX Design, GUI Applications)

Tools:

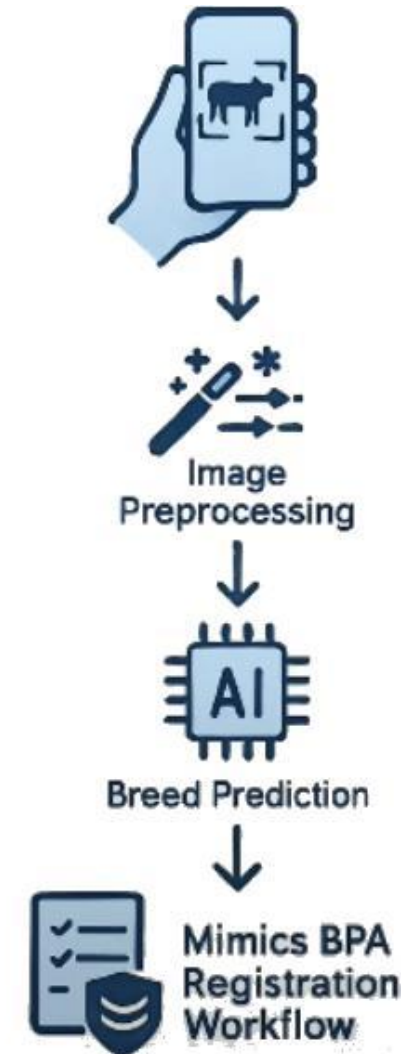
VS Code, Jupyter, Collab, GitHub, Figma, Git, Insomnia for API, Blender, Qt Design Studio

Inference Models:

Opencv + YOLOv8-Seg (You Only Look Once – Object Detection & Segmentation) + **Computer Vision (ResNet-50** (Deep Residual Networks for Image Classification)



Applications





FEASIBILITY AND VIABILITY

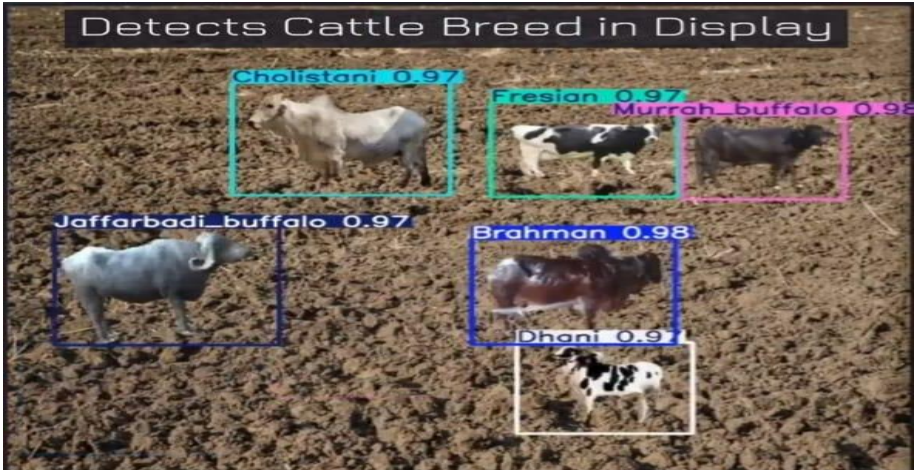


Feasibility:

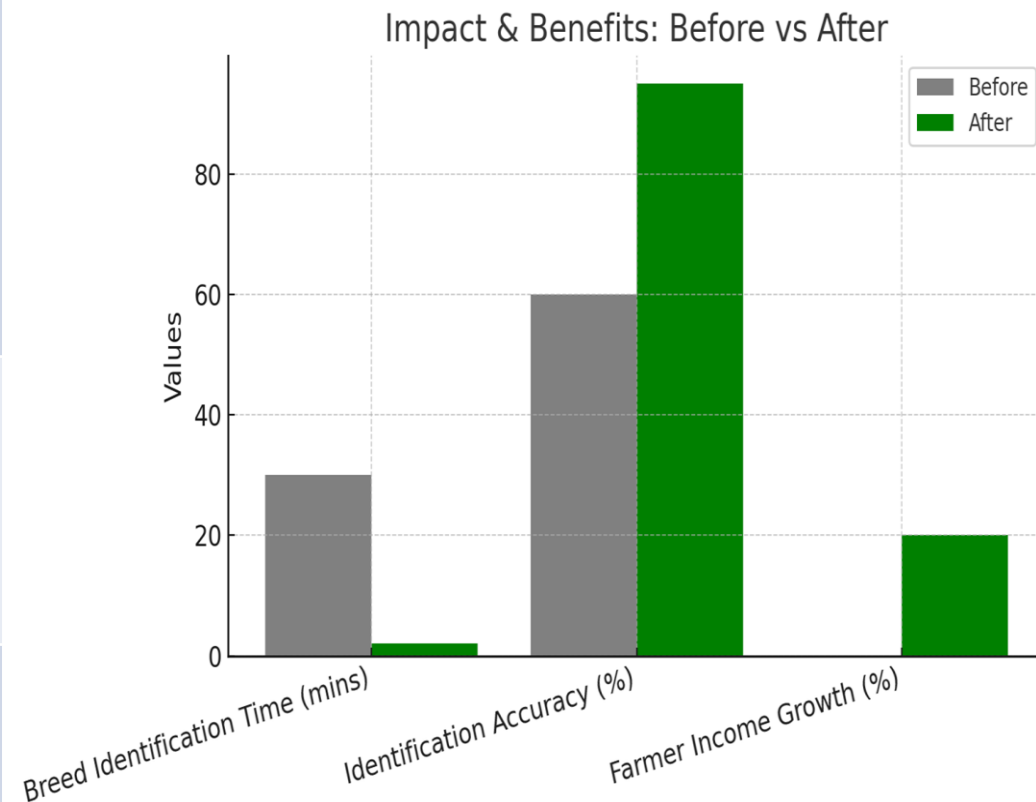
- **Team Expertise:** Python, Django, REST APIs, SQL, Qt Framework, PySide, UI/UX, Data Science, Deep Learning.
- **Proof of Concept:** Success of existing **object detection & image recognition systems** validates feasibility.

Challenges	Solutions
Data Diversity & Quality Issues – Breed images vary in lighting, pose, environment → impacts model accuracy.	Deploy Label Blend Studio for synthetic dataset generation, data augmentation, and auto-labeling to ensure high-quality, diverse training data.
Adoption Barriers for Field Workers (FLWs) – Limited technical literacy & low connectivity in rural areas.	Build offline-first mobile app with intuitive UI/UX, guided image capture, and automated feedback for easy adoption
Model Accuracy at Scale – Risk of misclassification in crossbred variations .	Implement continuous model retraining with real + synthetic datasets, active learning pipelines, and computer vision model optimization

Project Screenshots (Main):



Dimensions	Before(Manual System)	After (AI-Powered Solution)
Social Impact	❌ Breed misclassification, low trust in livestock records, limited farmer empowerment.	✅ AI/ML + Computer Vision ensures accurate livestock breed recognition , better animal welfare , breeding , healthcare , and farmer empowerment .
Economic Value	❌ Manual errors → cost overruns, inefficient subsidies, poor policy outcomes.	✅ Cost savings , efficient livestock registration , data-driven subsidies , improved productivity & resource allocation .
Scalability	❌ Fragmented systems, limited to specific regions.	✅ Cloud + Mobile deployment , scalable across indigenous & crossbred breeds , extendable to other livestock species & global markets .





भारत सरकार
GOVERNMENT
OF INDIA

- **FAO – Livestock Biodiversity Report**
 <http://www.fao.org/livestock-biodiversity/en/>
- **ICAR – Cattle & Buffalo Breeds in India**
 <https://icar.org.in/content/cattle-and-buffalo-breeds>
- **Government of India – Livestock Census (DAHD)**
 <http://dahd.nic.in/division-type/statistics-livestock-census>
- **Elsevier – Deep Learning for Livestock Identification (Computers & Electronics in Agriculture)**
 <https://doi.org/10.1016/j.compag.2020.105514>
- **Springer – Artificial Intelligence in Agriculture: Cattle Breed Recognition**
 <https://link.springer.com/journal/43681>
- **NDDB – National Dairy Development Board**
 <https://www.nddb.coop/services/animalbreeding>
- **Case Study – Dairy Co-ops**
 Field case studies on misclassification impact (milk yield pricing, income loss).